LESSON PLAN

GAS TURBINE ENGINE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	16
Total No. of Lesson Plans	16	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Calculation of engine thrust, various parameters and tests, adjustments of engine controls
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Calculation of engine thrust, various parameters and tests, adjustments of engine controls
5. Specific Objectives	13.1.8.1 Trainee will be able to calculate engine thrust

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Calculation of engine thrust using provided parameters; Safety: Verify data accuracy; Supervision: Instructor monitors calculation steps
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the definition of engine thrust?
(b) List three parameters required for thrust calculation.
(c) Name one test performed on engine controls.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Formula for calculating engine thrust • Variable definitions in thrust equation • Step-by-step calculation procedure • Unit conversion for thrust values • Sample problem solution	L & Ds	PPT, IB	15 min
(b)	• Engine performance test procedures • Control adjustment mechanisms • Interpretation of test data • Standard operating limits • Troubleshooting common deviations	L & Ds	PPT, PDF	20 min
(c)	• Final thrust verification steps • Documentation of test results • Adjustment of engine controls • Safety checks post-adjustment • Reporting anomalies	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how engine thrust is calculated using the derived formula.
(b) Describe the procedure for adjusting engine controls during a test.

12. Home Assignment and Follow Up	Prepare a brief summary of the thrust calculation method to be used in the next lesson on engine performance analysis.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min